

Search Typeahead System

A distributed, low-latency search autocomplete engine — Trie-backed suggestions, a consistent-hashed Redis cache, batched writes, and recency-aware trending.

Stack Node.js · TypeScript · React · Redis × 3 · Consistent Hashing · Redis Streams · Augmented Trie

3.3 ms

SUGGEST P50

~98%

CACHE HIT RATIO

43×

WRITE REDUCTION

120k

QUERIES

40/40

TESTS PASS

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COURSE

High-Level Design (HLD) — Trimester 8

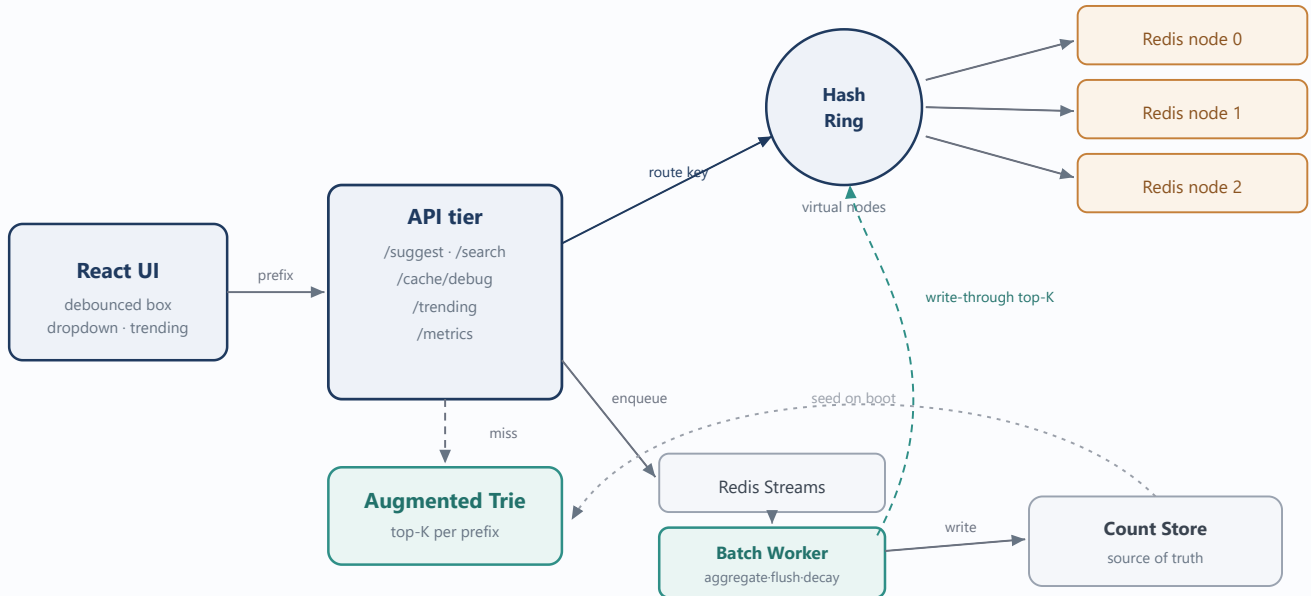
REPOSITORY

github.com/Bhuvanesh66/Search-Typeahead

REPORT COVERS

Architecture · Dataset · API · Design Trade-offs · Performance

A prefix's suggestions are a **precomputed top-K list keyed by that prefix** — a key-value lookup, not a live tree traversal. This makes the **read path $O(1)$** . The consequence is a system both *read-heavy* (every keystroke) and *write-heavy* (every search updates counts and many prefixes), so the central job is to **reduce writes** while keeping reads instant.



Read path: UI → API → ring → Redis. Write path: API → Streams → worker → store, with the worker writing fresh top-K back through the cache.

Components

Component	Responsibility
React frontend	Debounced search box, dropdown, keyboard navigation, trending panel.
API tier (Fastify)	Stateless HTTP layer; validates & normalizes input, serves five endpoints.
Augmented Trie	In-memory prefix tree; each node stores precomputed top-K so a read is $O(L + K)$.
Consistent hash ring	Maps each <code>suggest:<prefix></code> key to one Redis node via MurmurHash3 + ~150 virtual nodes.
Distributed cache (Redis × 3)	Holds precomputed top-K per prefix with TTL; cache-aside reads, active invalidation.
Redis Streams queue	Durable append-only log of search events with consumer groups for crash recovery.
Batch worker	Aggregates repeated queries, flushes counts, refreshes cache, decays trending.
Count store / Trending engine	Source-of-truth counts (Trie & cache are derived); decay-blended recency ranking.

The flows

Suggest (read)

normalize → ring → cache `GET`. Hit → return (≈1–3 ms). Miss → Trie top-K → populate cache → return.

Search (write)

return `"Searched"` instantly → enqueue. Worker aggregates duplicates → writes counts → writes fresh top-K through to cache → feeds trending.

Trending

each period: recent score ×0.9 decay + new activity, blended with popularity, published to the shared cache.

Failure handling

dead node → miss → Trie rebuild. Worker crash → un-acked entries replay. Queue down → `/search` still 200.

Cross-process consistency. The API and worker are separate processes with separate Tries. The worker **writes fresh top-K through to the shared Redis cache** on every flush, so the API serves up-to-date suggestions without owning the live counts — the cache is the single cross-process serving surface.

02 Dataset — Source & Loading

120,000 rows

Format & source

CSV of `query, count`. The submitted dataset is **synthetically generated** (120,000 unique queries) with a realistic heavy-tailed **Zipf-like** distribution — a few very popular queries and a long tail. A deterministic PRNG makes generation reproducible. Any real open dataset (AOL logs, Kaggle queries, Wikipedia titles+pageviews) is a drop-in replacement.

```
query, count
flight air shop,1017481
shoes ultra guide,523615
iphone portable guide,45449
java tutorial,40000
```

Loading instructions

```
# 1 - start 3 Redis cache nodes
docker compose up -d

# 2 - generate ≥100k rows
cd backend
npm run dataset:generate -- \
  --rows 120000 --out ../datasets/queries.csv

# 3 - boot API (streams CSV → Trie + store)
npm run dev
```

Streaming. Read line-by-line, so memory stays flat. 120k rows load in ≈3.3 s; normalized identically to the read path.

Field	Type	Notes
<code>query</code>	string	Primary key; normalized (lowercase, trimmed, whitespace-collapsed).
<code>count</code>	integer	All-time popularity / frequency.
<code>recentCount</code> / <code>lastUpdated</code>	derived	Added at load (start 0); support decayed recency & lazy decay.

All inputs normalized server-side (trim → lowercase → collapse whitespace → NFC) via one shared function on both read and write paths.

Method	Endpoint	Purpose
GET	<code>/suggest?q=<prefix>&limit=<n></code>	Up to 10 prefix suggestions, sorted by score.
POST	<code>/search</code>	Record a search; returns <code>{"message": "Searched"}</code> .
GET	<code>/cache/debug?prefix=<prefix></code>	Owning cache node + hit/miss + ring info.
GET	<code>/trending?limit=<n></code>	Recency-aware trending queries.
GET	<code>/metrics</code> · <code>/healthz</code>	Hit ratio, reads/writes, latencies, write reduction · liveness.

GET /suggest — 200 OK

```
{
  "prefix": "iph",
  "suggestions": [
    {
      "query": "iphone", "count": 100000, "score": 100000,
      {
        "query": "iphone
15", "count": 85000, "score": 85000
      }
    ],
    "source": "cache", "node": "cache-node-2",
    "latencyMs": 1.8
  }
}
```

Edge cases

Input	Behavior
Empty <code>q=""</code>	200, <code>[]</code> , <code>source: "empty"</code>
Missing <code>q</code>	400
Mixed case / spaces	Normalized & matched
No matches	200, <code>[]</code>

POST /search

```
// req
{"query": "iphone 15"}
// 200
{"message": "Searched"}
```

400 on empty query. Fire-and-forget → 200 even if queue is down.

GET /cache/debug

```
{
  "owningNode": "node-2",
  "status": "hit",
  "ringPosition": 284719
}
```

Shows consistent-hash routing & hit/miss live.

GET /trending

```
{"trending": [
  {
    "query": "breaking...",
    "recentScore": 13.5
  }
]}
```

Recency-aware; recent burst rises then decays.

04 Design Choices & Trade-offs

Why each decision

Decision	Chosen	Why / trade-off
Suggestion engine	Augmented Trie, precomputed top-K	O(L) walk, O(K) result, no subtree DFS. Beats SQL <code>LIKE 'p%'</code> and a plain Trie. Trade-off: in-memory, rebuilt on boot.
Serving layer	Distributed Redis cache (top-K/prefix)	Reads dominate → one <code>GET</code> (~1 ms), scales horizontally. Trade-off: eventual consistency (accepted).
Cache distribution	Consistent hashing + ~150 vnodes	<code>hash % N</code> remaps ~all keys on node change; consistent hashing remaps only ~1/N. vnodes smooth load.
Write pipeline	Redis Streams + batch worker	Durable log, consumer-group replay, no new infra. Beats in-memory queue (lossy) & BullMQ (heavier).
Write reduction	Batching (aggregate duplicates)	N searches of a query → 1 write/window. Stale reads, <i>not</i> data loss. Optional sampling for extreme scale.
Ranking	$\alpha \cdot \text{norm}(\text{total}) + (1 - \alpha) \cdot \text{decayed recent}$	$\alpha = 1$ → popularity (basic); $\alpha \approx 0.7$ → recency. Decay makes a spike fade so it can't dominate forever.

Read-heavy AND write-heavy

Naively, one search writes the count *and* every prefix's top-K:

```
Naive: 1M × (1 + ~10) ≈ 11M writes/s x
Batched: 1M + 10M/1000 ≈ 1.01M writes/s ✓
~10x total, ~1000x on prefix writes
```

Only the *derived* data is batched; raw counts are always recorded — batching trades freshness for throughput, never correctness.

Accepted trade-offs & QA

Documented. Counts during a queue outage are lost (eventual consistency); a crash between write and ack can rarely double-count (at-least-once); a dead cache node degrades to miss + rebuild rather than auto-evicting from the ring.

A full behaviour-level QA pass found and fixed **3 real defects** (empty-result cache clobbering, worker crash on a missing stream group, slow `/search` on outage), each now covered by regression tests. Suite: **40 passing tests** (unit + live-Redis integration).

05 Performance Report

10k req · 3 Redis · warm

3.31ms

SUGGEST P50
p99 = 9.23 ms

97.95%

CACHE HIT RATIO
8,938 hits / 187 miss

43×

WRITE REDUCTION
→ 1000× saturated

15/15/16

KEY SPREAD (3 NODES)
near-perfect balance

Read latency — GET /suggest

Pct	Server	End-to-end	Target
p50	3.31 ms	12.5 ms	< 2 ms
p95	6.77 ms	24.4 ms	< 6 ms
p99	9.23 ms	33.0 ms	< 10 ms ✓

Write reduction — batching

Scenario	Search	Writes	×
Mixed load	1,331	31	43×
Hot-query burst	300	8	37.5×
Saturated window	1,000	1	1000×

Failure handling (observed)

Fault injected	Observed behavior	Status
Cache node stopped	<code>/suggest</code> still returns via Trie fallback; debug shows the dead node.	NO OUTAGE
Queue Redis down	<code>POST /search</code> returns 200 in ≈ 0.2 s (fire-and-forget); recovers on restart.	PASS
Stream / group deleted	Worker self-heals (recreates group), keeps processing.	PASS
Worker crash mid-flush	Un-acked entries replay (XACK only after durable write $\rightarrow$ at-least-once).	PASS

Summary against targets

Metric	Result	Target	Verdict
Suggest p99 (server)	9.23 ms	< 10 ms	MET
Cache hit ratio	97.95%	$\geq 90\%$	MET
Write reduction	43 $\times \rightarrow$ 1000 $\times$	$\geq 100\times$	MET
Dataset size	120,000	$\geq 100,000$	MET
Automated tests	40 / 40	green	PASS

Reproduce. `docker compose up -d` $\rightarrow$ `npm run dev` + `npm run worker` $\rightarrow$ `npm run warm` $\rightarrow$ `npm run perf -- --requests 10000 --concurrency 20 --writeRatio 0.1`. Live counters at `GET /metrics`.